

Measuring Thematic Alignment and Drift in AI Journals: A Sentence-BERT and BERTopic Study of ESWA and KBS (2015–2025)

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Abstract

Scientific journals define their academic identity through their Aims & Scope statements, but the themes reflected in published papers may shift over time due to evolution of research field. This project investigates whether articles published in two AI-related journals, *Expert Systems with Applications* (ESWA) and *Knowledge-Based Systems* (KBS), remained thematically aligned with their journal scopes between 2015 and 2025.

The study uses 5,788 cleaned article abstracts collected from Semantic Scholar. Sentence-BERT embeddings were used to represent journal scopes and article abstracts, and cosine similarity was used as the main thematic alignment score. BERTopic was applied to identify major thematic areas, while statistical trend analysis and a TF-IDF Logistic Regression classifier were used to examine thematic drift and journal-level distinguishability.

The results show a weak but statistically significant decrease in thematic alignment over time for both journals. ESWA showed slightly higher average alignment than KBS, although the practical difference was modest. Topic modeling and outlier inspection suggest that lower-alignment papers were often related to newer or more specialized AI areas rather than being irrelevant. Overall, the project shows how NLP methods can be used to the study of thematic coherence, drift, and editorial focus in scientific publishing.

Keywords: thematic alignment, scientific journals, Sentence-BERT, BERTopic, topic modeling, semantic similarity, TF-IDF, Artificial Intelligence

1 Introduction

Scientific journals define their academic identity through their Aims & Scope statements. These statements describe the topics, methods, and types of contributions that a journal intends to publish. However, research fields change over time, and this is especially visible in Artificial Intelligence. Earlier AI research was often connected with expert systems, knowledge representation, symbolic reasoning, and decision support, while recent AI research increasingly includes machine learning, deep learning, computer vision, natural language processing, and privacy-preserving methods.

This project studies whether articles published in AI-related journals remain thematically align to their journal scopes. The basic idea is to compare the semantic meaning of each article abstract with the semantic meaning of the journal’s Aims & Scope. Higher semantic similarity indicates stronger thematic alignment, while lower similarity may mean thematic drift or the presence of newer and more specialized research areas.

The study focuses on two journals: *Expert Systems with Applications* and *Knowledge-Based Systems*. Both journals are related to Artificial Intelligence, intelligent systems, decision support, and knowledge-based methods, but they also publish a wide range of applied and technical research. This makes them suitable case studies for analyzing thematic alignment and drift.

The aim of the project is not to judge whether individual papers should or should not have been published. Instead, the aim is to measure thematic proximity and interpret general patterns. A low alignment score does not necessarily mean that a paper is outside the journal’s scope; it may simply indicate that the paper uses newer terminology, focuses on a specialized application, or belongs to an emerging subfield that is less directly represented in the scope statement.

2 Research Question and Methodology

2.1 Research Question

The main research question of this project is:

How has thematic alignment between published articles and journal Aims & Scope evolved in selected Artificial Intelligence journals between 2015 and 2025?

The project also considers three questions:

1. Is there evidence of thematic drift over time in the selected journals?
2. Which topics are most and least aligned with the journal scopes?
3. Are the two journals thematically distinguishable based on their article abstracts?

2.2 Measurable Objectives

The project has the following measurable objectives:

- Collect and clean article abstracts from two AI-related journals for the period 2015–2025.
- Represent each journal scope and each article abstract using semantic embeddings.

- Compute an alignment score for every article using cosine similarity.
- Analyze yearly changes in alignment scores to find possible thematic drift.
- Use BERTopic to identify major thematic areas in the corpus.
- Compare the two journals using statistical tests.
- Train a classifier baseline to test whether the journals are distinguishable from abstracts.

2.3 Formal Problem Definition

Let each journal be represented by an Aims & Scope text S_j , where j identifies the journal. Each article i published in journal j is represented by its abstract A_i . The goal is to measure the semantic proximity between the article abstract A_i and the corresponding journal scope S_j .

A sentence embedding model f is used to map both texts into a shared vector space. The abstract embedding is defined as:

$$v_i = f(A_i)$$

The journal scope embedding is defined as:

$$s_j = f(S_j)$$

The thematic alignment score for article i is then calculated as the cosine similarity between the article abstract embedding and the journal scope embedding:

$$Alignment(A_i, S_j) = \cos(v_i, s_j)$$

A higher score indicates stronger semantic proximity between the article abstract and the journal scope. A lower score indicates weaker semantic proximity. The score is then analyzed by journal, year, and topic.

2.4 Dataset and Preprocessing

The dataset was collected from Semantic Scholar. The collected metadata includes article titles, abstracts, publication years, journal names, paper identifiers, URLs, citation counts, and related fields of study.

The analysis focuses on articles published between 2015 and 2025 in two journals:

- *Expert Systems with Applications* (ESWA)
- *Knowledge-Based Systems* (KBS)

The raw data was cleaned before analysis. Papers without abstracts were removed because the project depends on abstract-level semantic comparison. Duplicate paper IDs were removed to avoid counting the same article more than once. Very short abstracts were also excluded because they may not contain enough information for reliable semantic analysis.

After preprocessing, the final cleaned dataset contained 5,788 papers: 3,465 from ESWA and 2,323 from KBS.

2.5 Sentence-BERT Alignment Method

For each journal, the Aims & Scope text was saved as a reference document. Each article abstract was then compared with the corresponding journal scope.

Sentence-BERT was used to convert both the journal scope texts and the article abstracts into dense semantic embeddings. This approach is suitable because it compares texts based on meaning rather than only exact word overlap. The main alignment metric is cosine similarity. However, this metric should be interpreted carefully because it measures semantic proximity to the scope text, not the academic quality or publication suitability of the article.

2.6 Topic Modeling and Statistical Analysis

BERTopic was used to identify major thematic areas in the dataset. A single topic model was trained using abstracts from both journals, which made it possible to compare the same topic structure across ESWA and KBS.

To examine thematic drift, alignment scores were summarized by journal and year. Spearman correlation was used to measure whether alignment scores generally increased or decreased over time. Linear regression was used to estimate the direction and strength of the yearly trend. A Mann-Whitney U test was used to compare the alignment score distributions of ESWA and KBS.

As an additional experiment, a TF-IDF and Logistic Regression classifier was trained to predict whether an abstract came from ESWA or KBS. This classifier was included as a baseline to test whether the two journals are thematically distinguishable using abstract text alone.

3 Experimental Results

3.1 Dataset and Evaluation Metrics

The final cleaned dataset contains 5,788 papers: 3,465 from ESWA and 2,323 from KBS. Papers without abstracts, duplicate paper IDs, and very short abstracts were removed before analysis.

Table 1 Number of cleaned papers per journal.

Journal	Short name	Number of papers
<i>Expert Systems with Applications</i>	ESWA	3,465
<i>Knowledge-Based Systems</i>	KBS	2,323
Total		5,788

The main evaluation metric is the alignment score, calculated as cosine similarity between the Sentence-BERT embedding of an article abstract and the embedding of

the corresponding journal Aims & Scope. The analysis also uses yearly mean alignment scores, Spearman correlation, linear regression, Mann-Whitney U test, BERTopic topic-level alignment, and classifier performance metrics.

3.2 Alignment and Thematic Drift

The overall alignment scores show that ESWA has slightly higher average alignment than KBS. The mean alignment score for ESWA is 0.120, with a median of 0.111. For KBS, the mean alignment score is 0.106, with a median of 0.098. This suggests that ESWA articles are slightly closer to their journal scope according to the embedding-based metric, although the practical difference is modest.

The trend analysis shows a weak negative relationship between publication year and alignment score for both journals. For ESWA, the Spearman correlation is approximately -0.193. For KBS, it is approximately -0.156. In both cases, the p-values are very small, indicating that the trends are statistically significant.

Table 2 Trend statistics.

Journal	Spearman	Spearman p	Slope	Linear p	R^2
ESWA	-0.193	2.58e-30	-0.0094	4.15e-33	0.041
KBS	-0.156	4.47e-14	-0.0075	5.29e-14	0.024

The linear regression results also show negative slopes for both journals. This means that, on average, alignment scores slightly decrease over time. However, the R-squared values are low, meaning that publication year explains only a small part of the variation in alignment scores. Therefore, the result should not be overstated. The evidence suggests gradual thematic drift, but not a drastic change from the journals' scopes.

3.3 Topic Modeling and Classifier Baseline

BERTopic identified nine main topics in the dataset. The most aligned topic was related to process systems, event systems, and software, with a mean alignment score of approximately 0.259. Other highly aligned topics included fuzzy decision-making, rule-based knowledge discovery, knowledge modeling, recommendation systems, and clustering. The least aligned topics included broader machine learning models, optimization algorithms, privacy-preserving learning, and federated learning.

This result helps explain the observed thematic drift. The decrease in alignment over time may be partly caused by the rise of newer AI subfields whose terminology differs from the wording used in the original scope statements. These topics are not outside AI, but their vocabulary is less directly connected to the traditional language of expert systems and knowledge-based systems.

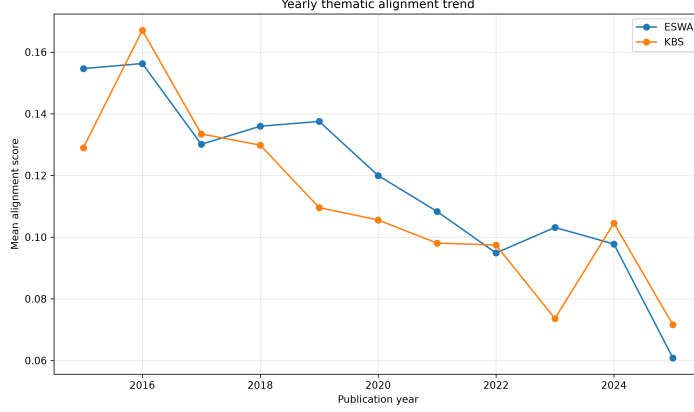


Fig. 1 Yearly alignment trend.

The TF-IDF and Logistic Regression classifier achieved an accuracy of approximately 69.2% when predicting whether an abstract belonged to ESWA or KBS. This is higher than the majority-class baseline of approximately 59.9%.

Table 3 Classifier performance.

Class	Precision	Recall	F1-score	Support
ESWA	0.754	0.720	0.737	693
KBS	0.609	0.649	0.629	465
Accuracy			0.692	1,158
Macro average	0.681	0.685	0.683	1,158
Weighted average	0.696	0.692	0.693	1,158

The classifier result suggests that the two journals are thematically distinguishable to some extent. However, the accuracy is moderate rather than high, which is expected because both journals overlap significantly in their focus on Artificial Intelligence, intelligent systems, decision support, and knowledge-based methods.

3.4 Outlier Analysis

The outlier inspection provides a qualitative check on the alignment scores. High-alignment papers often contain terms directly related to the journal scopes, such as expert systems, knowledge acquisition, decision models, fuzzy logic, knowledge configuration, and knowledge graphs. In contrast, low-alignment papers often focus on specialized technical subjects such as generative adversarial networks, biomedical image segmentation, code-switched emotion detection, video colorization, convolutional neural networks, and binary sequence optimization.

These topics are still related to Artificial Intelligence, but their abstracts are less directly connected to the wording of the journal Aims & Scope. This confirms that the alignment score is useful, but it also has limitations. Low alignment should be interpreted as lower semantic proximity, not as proof that a paper is unsuitable for the journal.

4 Conclusion

The results indicate that thematic alignment in both journals has slightly declined over time. This supports the idea that Artificial Intelligence has expanded into many new technical and application-driven areas. The trend is statistically significant, but weak, so it should be interpreted carefully rather than as a strong deviation from the journals' scopes.

The topic modeling results help explain this pattern. Topics related to knowledge-based systems, decision support, fuzzy decision-making, and rule-based systems show higher alignment. In contrast, newer areas such as privacy-preserving learning, federated learning, deep learning-based segmentation, and advanced optimization show lower alignment. This does not mean that these topics are irrelevant; rather, their vocabulary and focus may be less directly represented in the journal scope statements.

The classifier baseline also shows that the two journals are thematically distinguishable to some extent, although there is considerable overlap between them. Overall, the project shows that NLP methods can be used to quantify thematic coherence, identify drift, and inspect outlier papers in scientific publishing. At the same time, the results should be interpreted critically because alignment scores measure semantic proximity, not academic relevance or publication quality.

4.1 Limitations

The main limitations are that the analysis is based only on article abstracts, uses the current Aims & Scope text for the whole period, and depends on the quality of Semantic Scholar metadata. Future work could use full-text articles, historical scope statements, more journals, and different embedding models to test whether the findings remain stable.

AI Usage Disclaimer

Parts of this project were developed with the assistance of OpenAI's ChatGPT (GPT-5.5). The AI was used to support the workflow structuring, code explanation, debugging, drafting of descriptive texts, and improving readability. All content produced with AI assistance has been carefully reviewed, edited, and validated by me. I take full responsibility for the final content and its accuracy, relevance, methodology, interpretation, and academic integrity.

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